Predicting How Fast Bangkok City Complaints Get Fixed: A Data Mining Study of Traffy Fondue

Team	Aung Myo (6702955), Aye Chan Ko (6703266), Phoo Myat May Zaw (6703013)
School	Rangsit University International College · Academic Year 2026
Track	Classification (main) + Clustering
Status	Data is public. Official CSV download from Traffy (BMA) is confirmed.

1. Problem Statement

Traffy Fondue is an app (on LINE) where Bangkok people report city problems, like broken roads, flooding, street lights and rubbish. It has more than **one million** fixed complaints [1]. Some problems are fixed in a few days, but others wait for weeks. Right now, city staff have **no tool** to tell early which new complaints will be slow.

In other countries, machine learning already helps sort and route public complaints [2]. On this data, past work mostly sorts complaint text into the right *category* [3]. One student project, Sed-Young-Wa [4], tried to predict fixing time with a single regression model (error RMSE = 20). We found **no published study** that compares several models, explains **why** complaints are slow, or groups districts by fixing speed. This is the gap our project fills, and we will use [4] as a baseline to beat.

2. Research Questions

- **RQ1.** Which factors best predict if a complaint is fixed **fast** (within a few days, 3 or 7, chosen from the data) or **slow**? For example: problem type, district, responsible office, time and day reported, report made outside office hours, photo attached.
- **RQ2.** Which model works best: Logistic Regression, Decision Tree, Random Forest or XGBoost? Does it beat the earlier Sed-Young-Wa result [4], and does it still work well on the newest data (Jan–Oct 2025)?
- **RQ3.** Can Bangkok's 50 districts be grouped by their problem types and fixing speed? Which groups are always slow?

3. Dataset

- **Source:** Traffy Fondue open data from the Bangkok Metropolitan Administration (BMA), free CSV download at publicapi.traffy.in.th/dump-csv-chadchart/bangkok_traffy.csv
- **Size:** over one million real records (about 25,000–35,000 per month), far more than the 500-record minimum. Thai data, which gives higher novelty.
- **Study period:** June 2022 – October 2025. The problem type field is empty from November 2025 on (100% empty in 2026), so later data is left out. Rows with no type are removed.
- **Main fields (checked):** type, comment, photo, coords, district, subdistrict, organization, state, timestamp, last_activity, star (citizen rating), count_reopen (times reopened).
- **Target:** fixing time = last update date – created date. Only finished cases (state = finished) are used. We turn it into two classes: **Fast** or **Slow**.
- **New variables not used before:** responsible office, times reopened, photo attached. Plus daily rainfall and holidays.
- **Known limits:** last update time is not always the real fix date (some cases are updated months later), so extreme cases will be checked. Citizen rating is mostly empty.
- **Privacy:** the public data has no personal names or phone numbers.

4. Method (CRISP-DM)

Step	What we will do
Understand data	Look at problem types, districts and time patterns with charts.
Prepare data	Remove duplicates and unfinished cases, fix missing values, create time features (hour, weekday, holiday), balance the classes on training data only.
Build models	Logistic Regression (baseline), Decision Tree, Random Forest, XGBoost, tuned with 5-fold cross-validation. K-means and hierarchical clustering for districts.
Evaluate	Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrix. Train on 2022–2024, test on 2025. Silhouette score for clusters.
Explain	Feature importance and SHAP to show **why** a complaint is slow. Map of slow districts.

5. Expected Contribution

- First study to **compare several models** for fast/slow fixing on Traffy Fondue, and to **explain** the results with SHAP, using newer data up to Oct 2025, which past work did not cover.
- First grouping of Bangkok districts by complaint mix and fixing speed.
- Helps the BMA spot slow complaints early and see which districts and offices need more help.
- Clean, commented code and a repeatable pipeline, ready for a conference paper.

6. Timeline

Milestone	Deliverable	Due
#1	Team consult, idea pitch, printed proposal	Week 8 (14–16 Oct 2026)
#2	Dataset confirmed, 1-page proposal	Week 10
#3	Data cleaned, first results	Week 12
#4	Final presentation, IEEE paper, code and dataset	Week 14

References

[1] BMA Public Relations Office, "Bangkok Celebrates Digital Milestone as Traffy Fondue Resolves One Million City Complaints," Mar. 2026. https://pr-bangkok.com/?p=583017

[2] F. Caldeira, L. Nunes and R. Ribeiro, "Classification of Public Administration Complaints," SLATE 2022, OASIcs vol. 104, 2022. doi:10.4230/OASIcs.SLATE.2022.9

[3] A. Pongsathornwiwat and P. Kaewvichit, "Developing Multi-Label Classification Model for Improving Text Categorizing Problems a Case of Traffy* Fondue Platform," *Science & Technology Asia*, vol. 29, no. 2, pp. 138–147, 2024. doi:10.14456/scitechasia.2024.32

[4] karnjj, NonKhuna et al., "Sed-Young-Wa: A service for estimating the end of the issues submitted in the Traffy Fondue platform," GitHub repository, 2023. https://github.com/karnjj/sed-young-wa

A full literature review (at least 10 references from the last 5 years) will be done for the final paper. References above were read at abstract level.